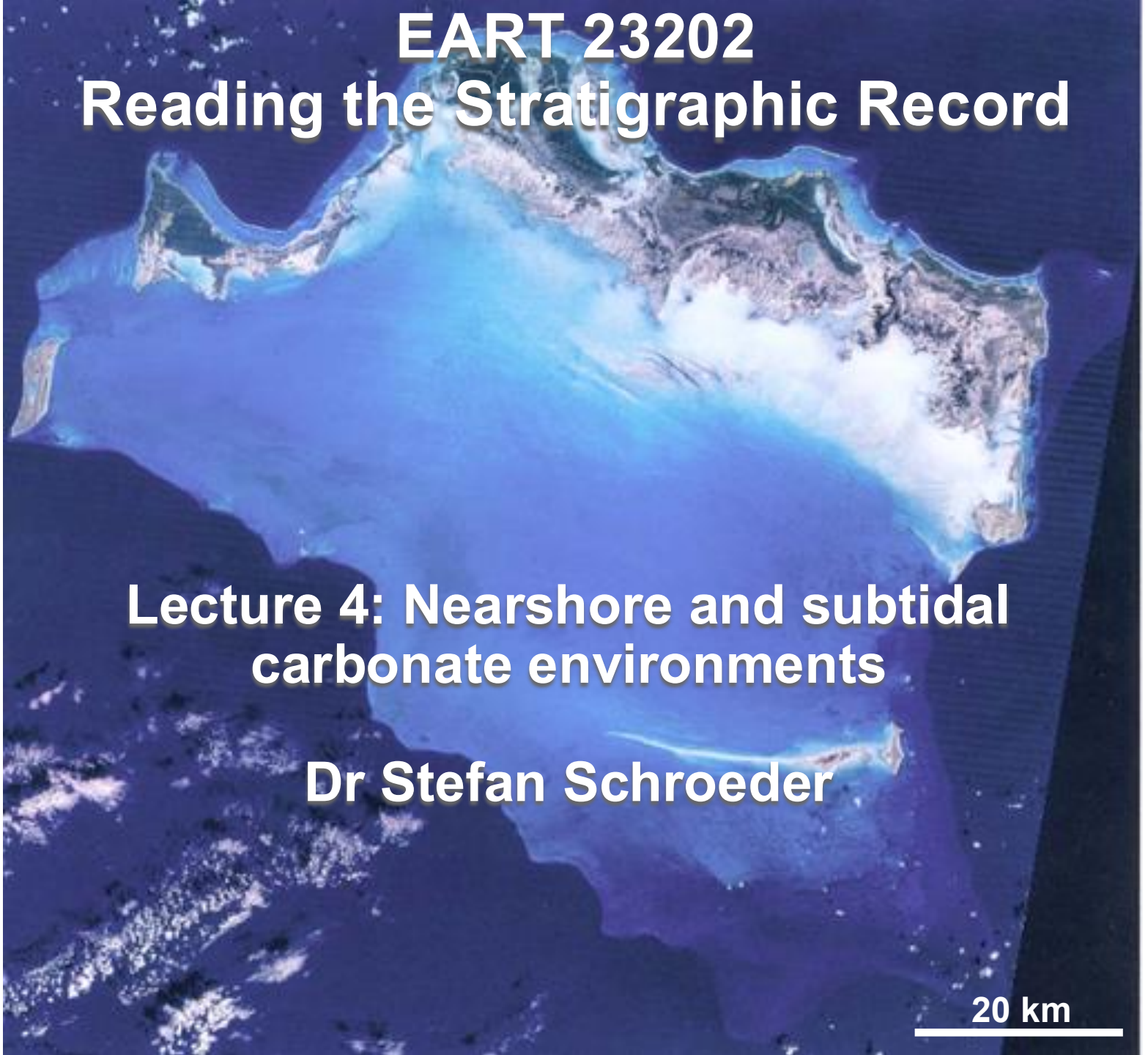
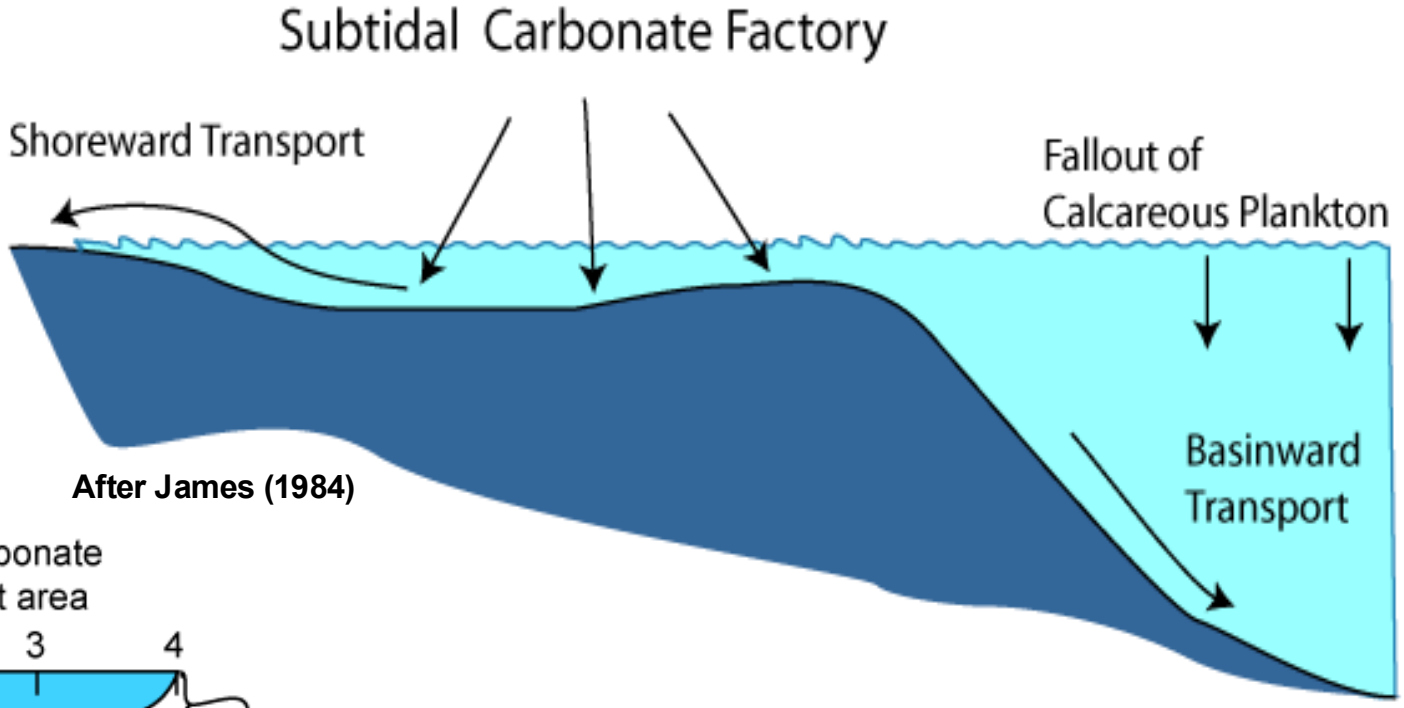
EART 23202
Reading the Stratigraphic Record



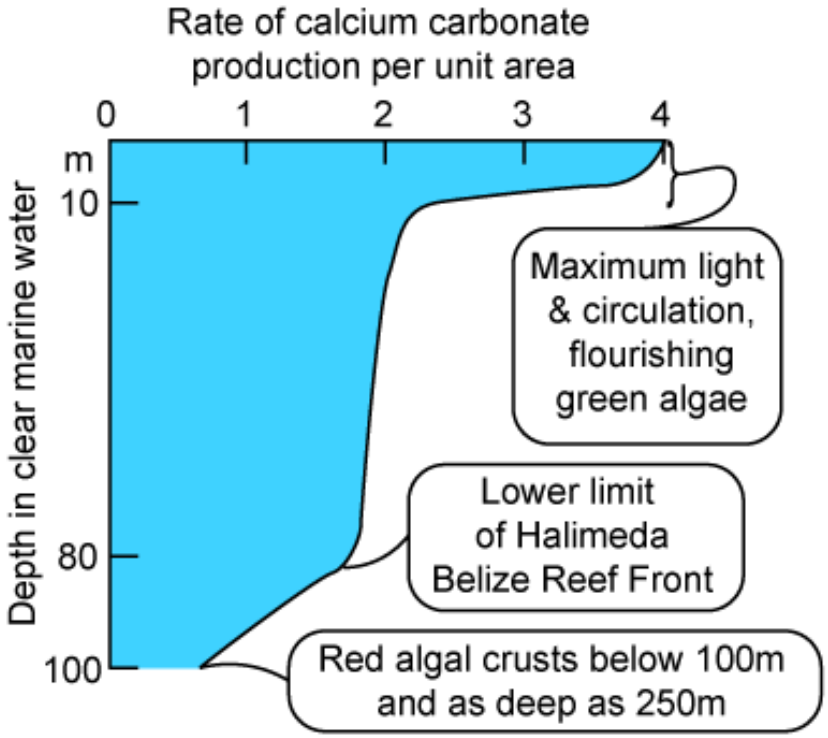
Lecture 4: Nearshore and subtidal carbonate environments

Dr Stefan Schroeder

Carbonate factory

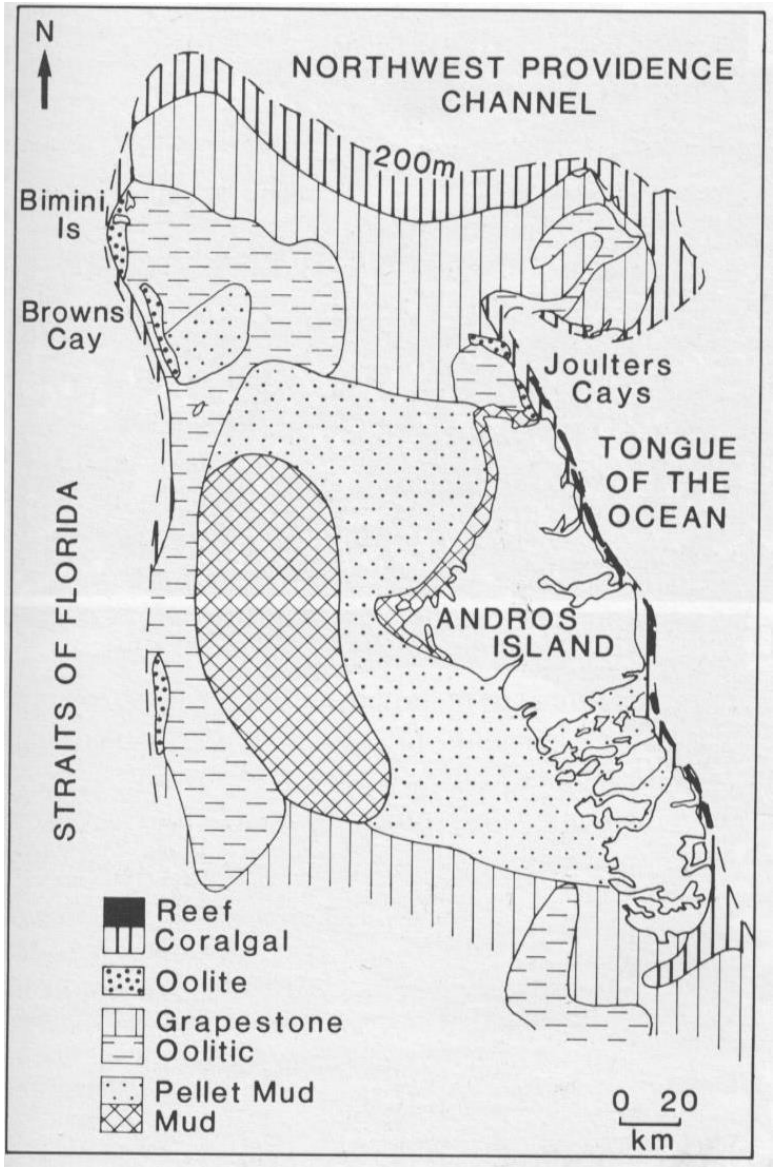


After James (1984)



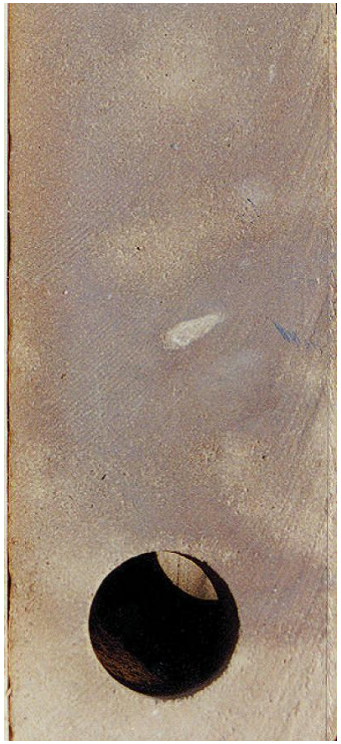
- **Carbonate production is largely bio-chemical, favored by:**
  - **Light**
  - **Nutrients**
  - **Warm water**
  - **Elevated salinity**
  - **Protection from terrigenous influx**
- **Production maximal in shallow, (sub) tropical areas: carbonate shelves**

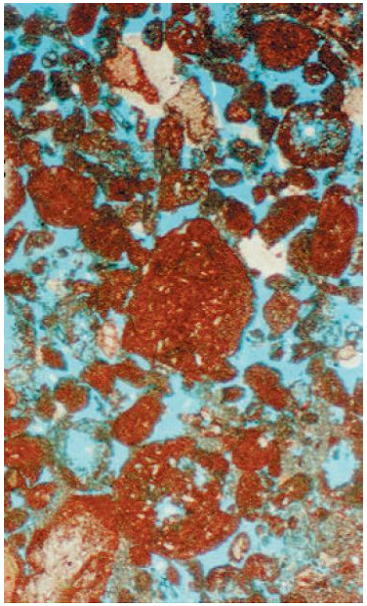
Depositional environments + lithofacies



Tucker, 1990

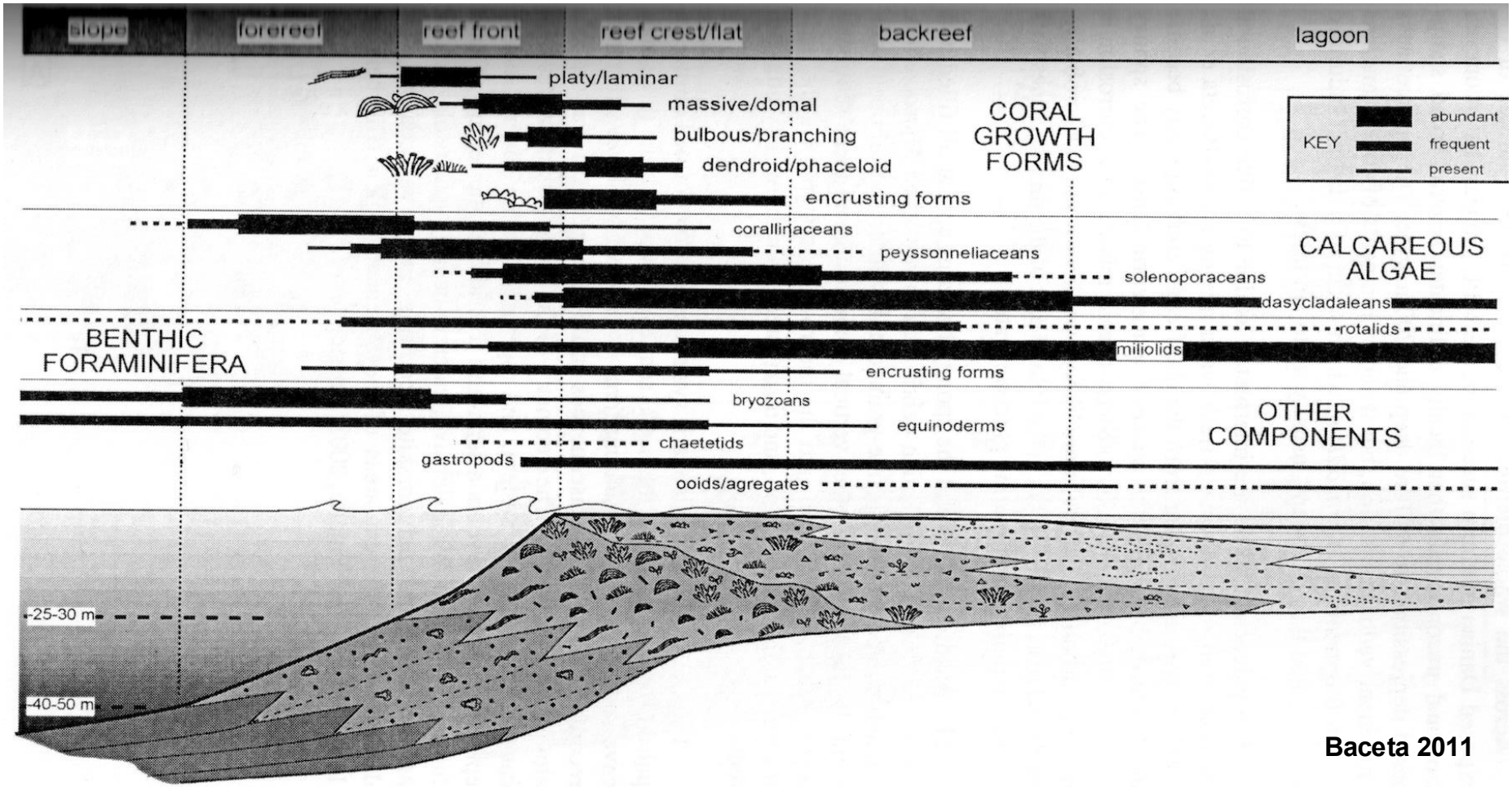
- Distribution of modern sediments reflects physical environment: water depth, nutrient supply, energy
- Lithofacies: biogenic & abiogenic sediment composition
- Facies analysis (macro- and microfacies): reconstruct depositional environments & processes





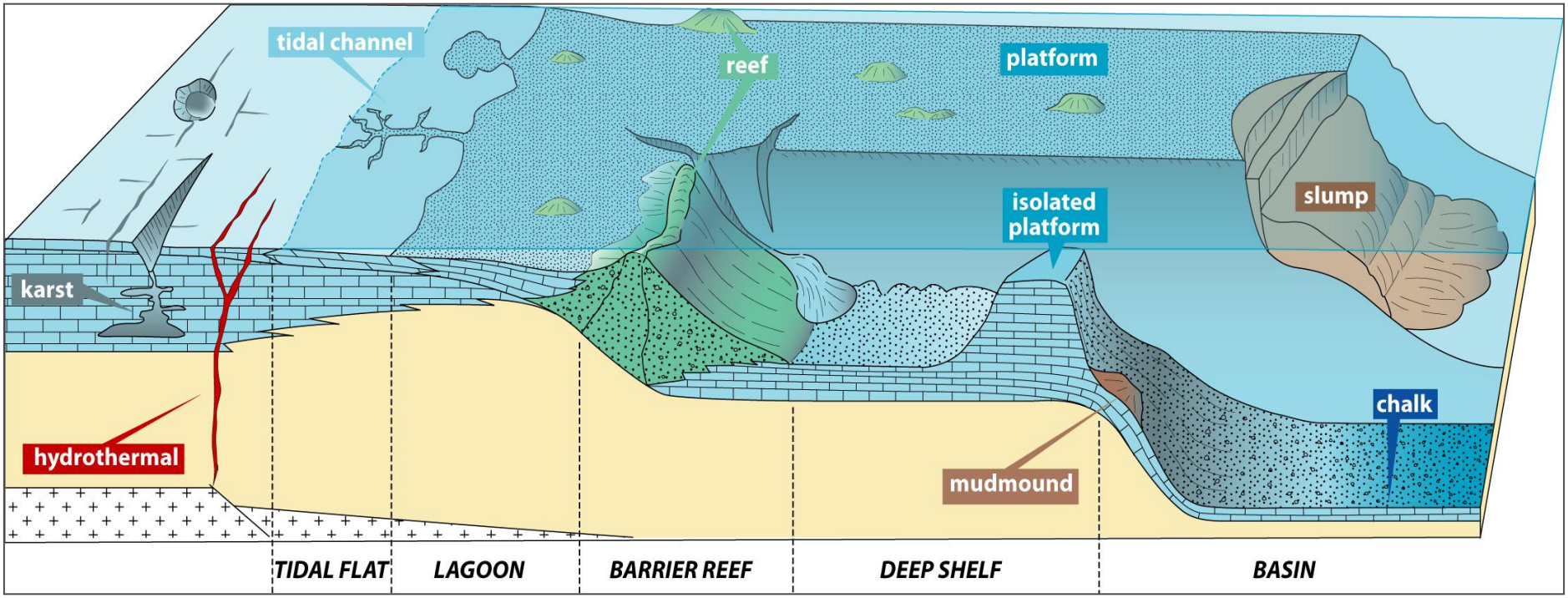
Skeletal grainstone
Algal peloidal grainstone

Carbonate depositional environments



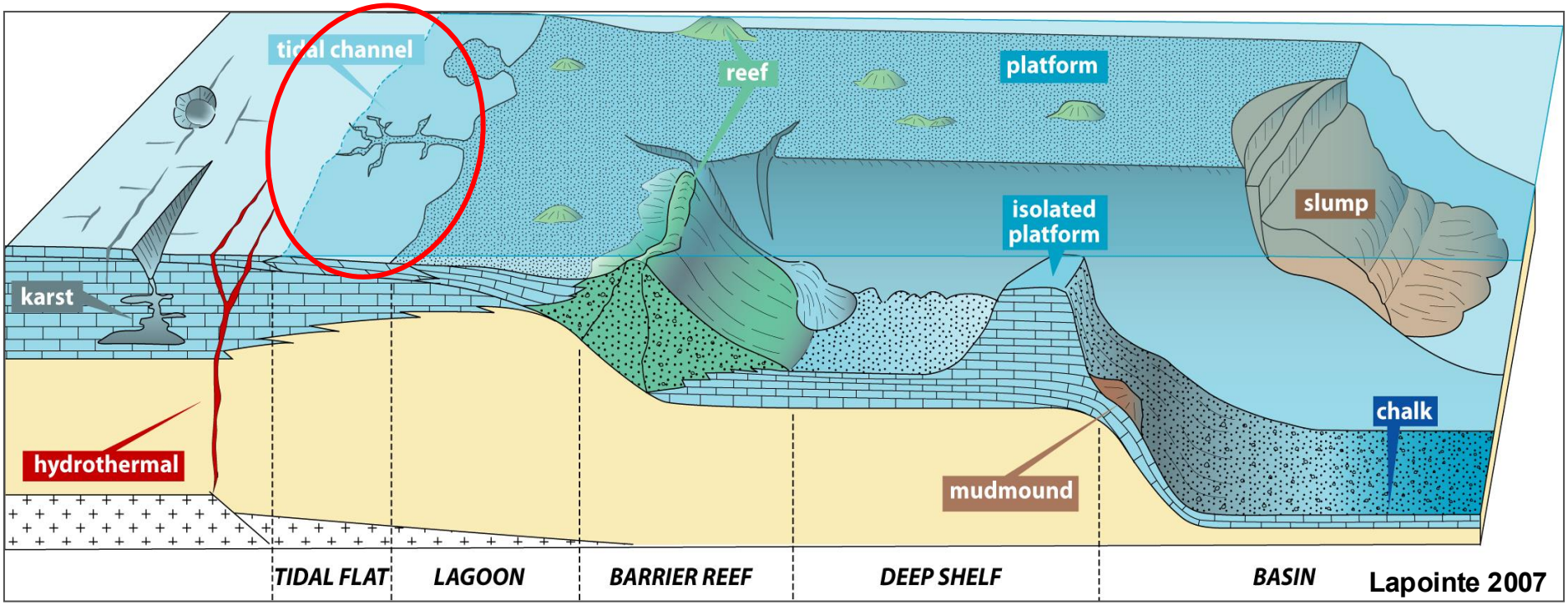
Baceta 2011

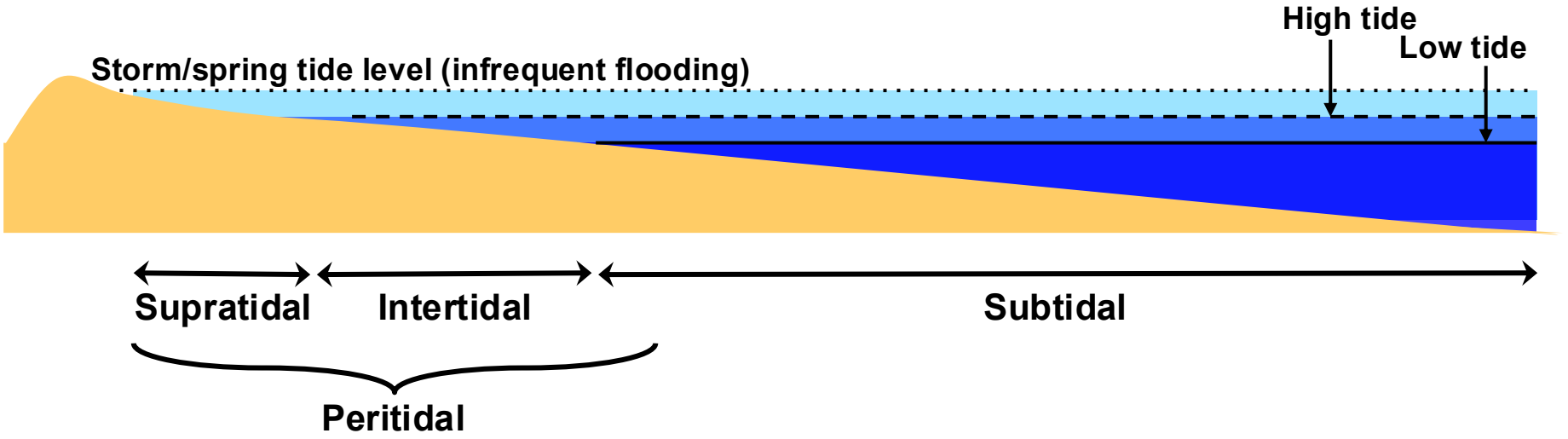
Carbonate depositional environments



Lapointe 2007

Carbonate depositional environments



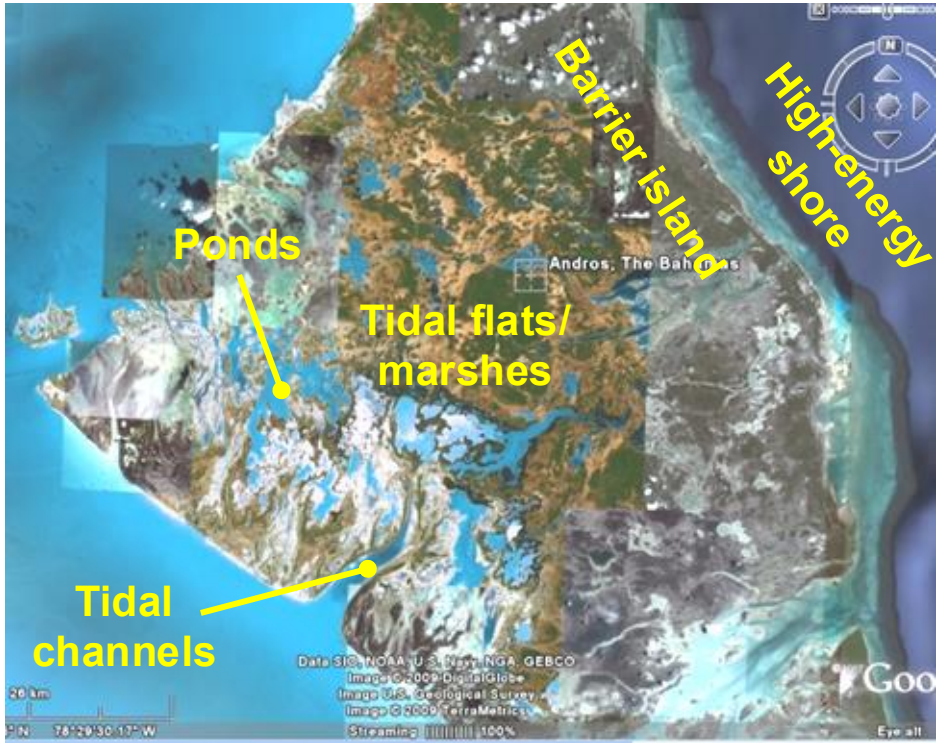


Peritidal facies





- most peritidal facies develop where tidal range is low (<2m)
- different facies mosaics and diagenesis/hydrology under humid and arid conditions
- topography can influence development of islands and lagoons
- sediment source: subtidal
- channels rework sediment and focus onshore sediment transport during storms
- low faunal diversity



Peritidal facies
Supratidal zone


Desiccation cracks

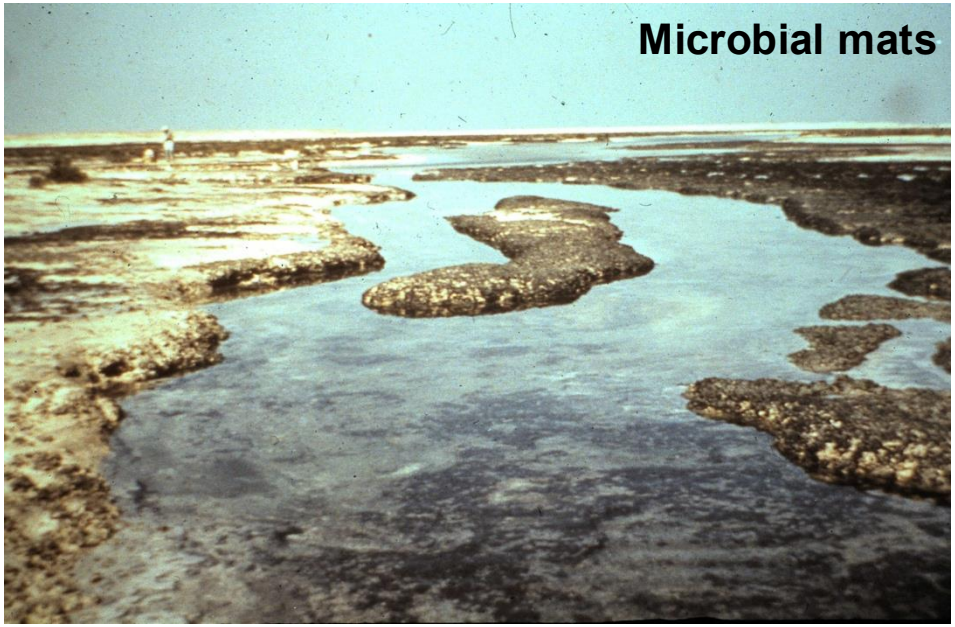


Flat-pebble conglomerate

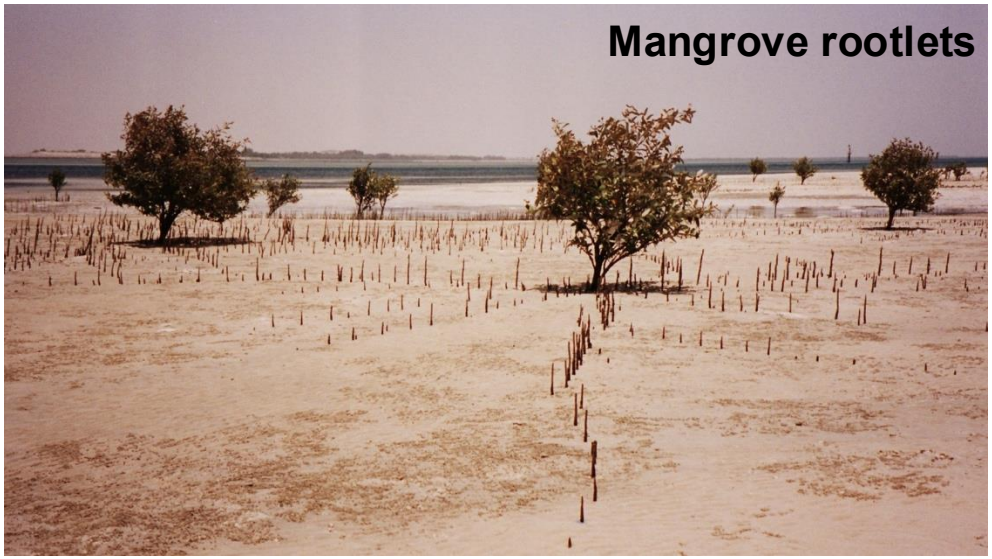
- Presence of plant debris, rootlets (mangroves), algal mats
- Mostly muddy sediment
- Depositional processes and sedimentary structures:
  - Desiccation cracks
  - Reworking of lithified crusts by storms: mud chips, flat-pebble conglomerates
  - Fenestrae


Fenestrae

Peritidal facies
Intertidal zone

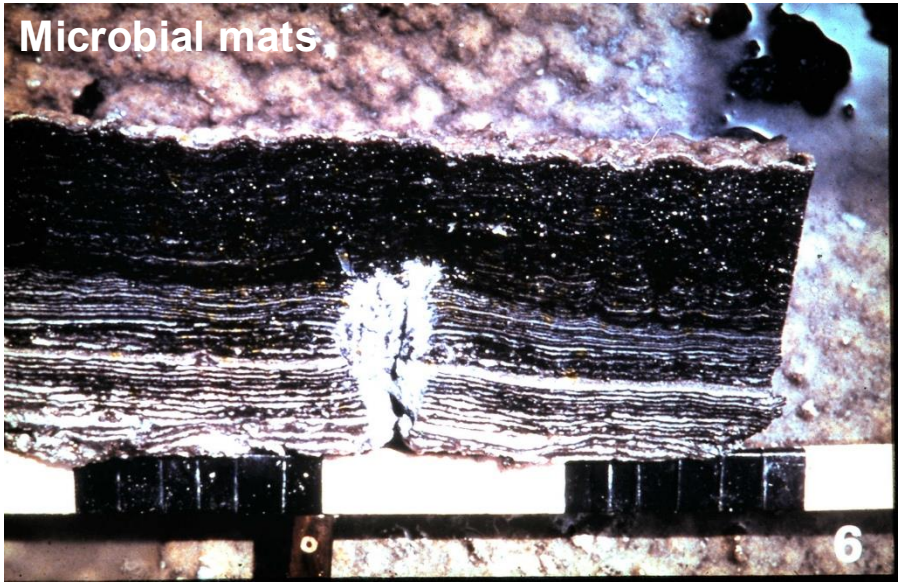




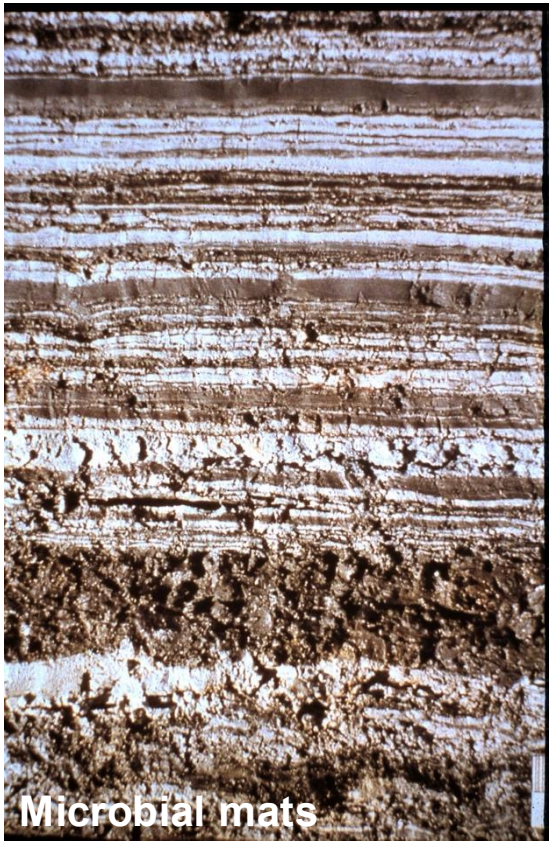




Peritidal facies
Microbial mats & stromatolites



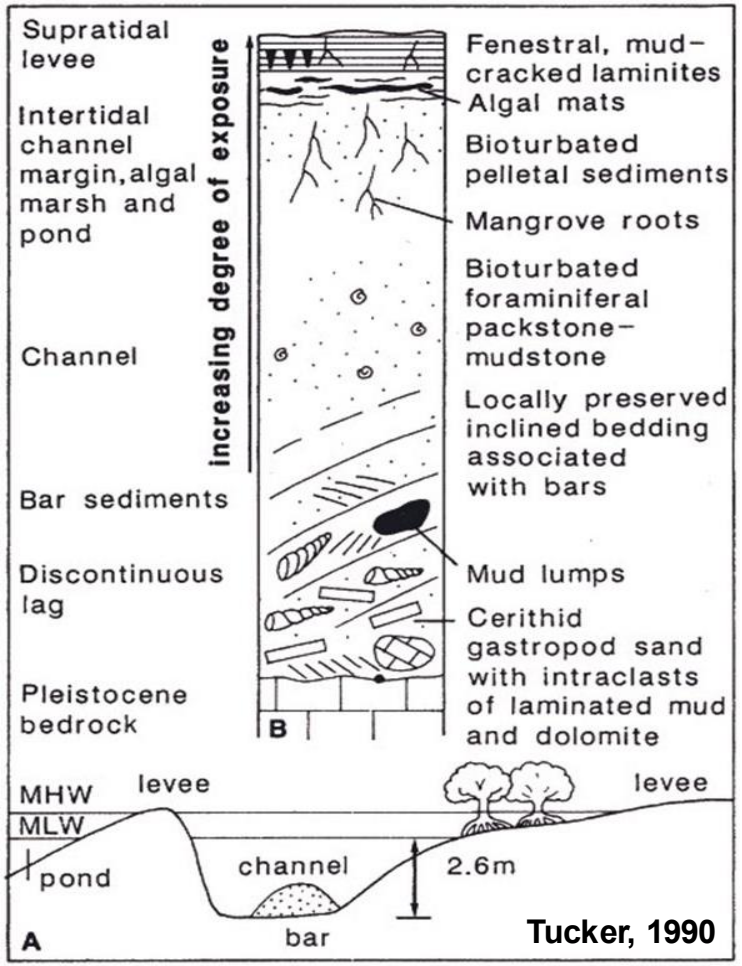




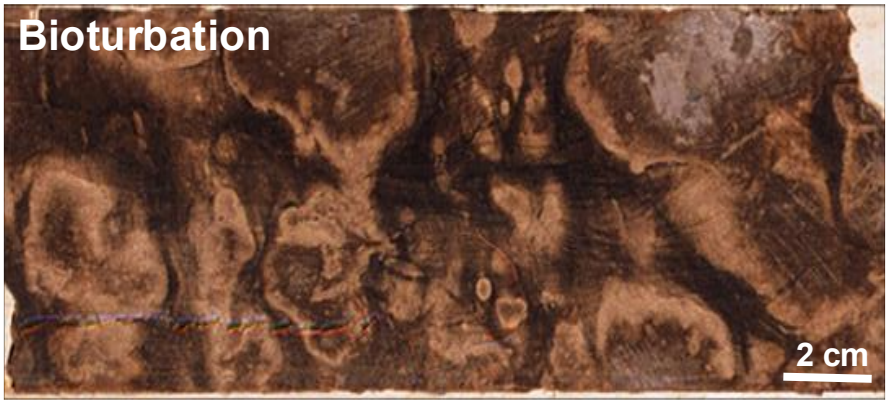
- Presence of rootlets (mangroves)
- Low-diversity, high-abundance skeletal fauna
- Mostly muddy sediment
- Depositional processes and sedimentary structures:
  - Desiccation polygons, fenestrae
  - Crinkly (microbial) lamination
  - Bioturbation in lower intertidal

Peritidal facies
Subtidal zone

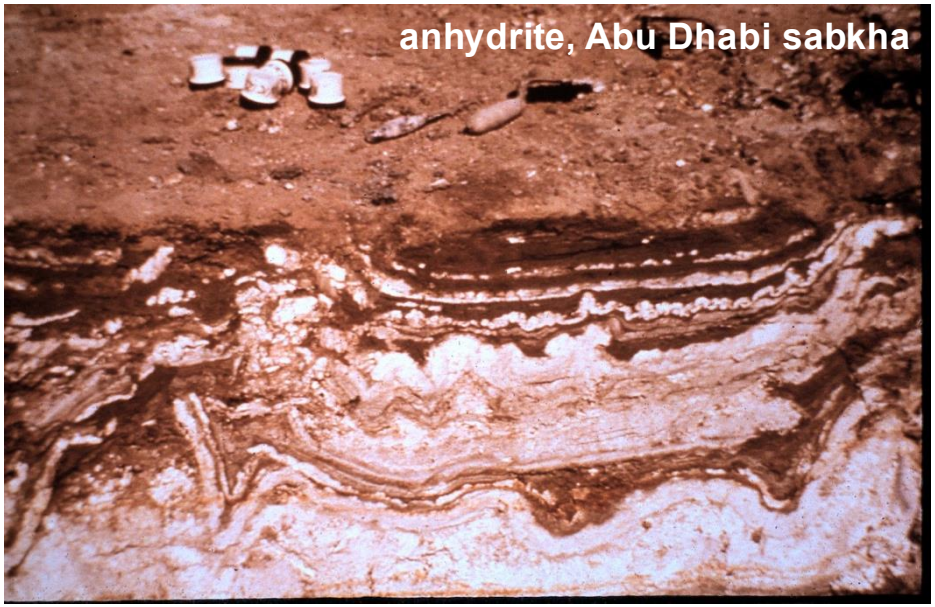


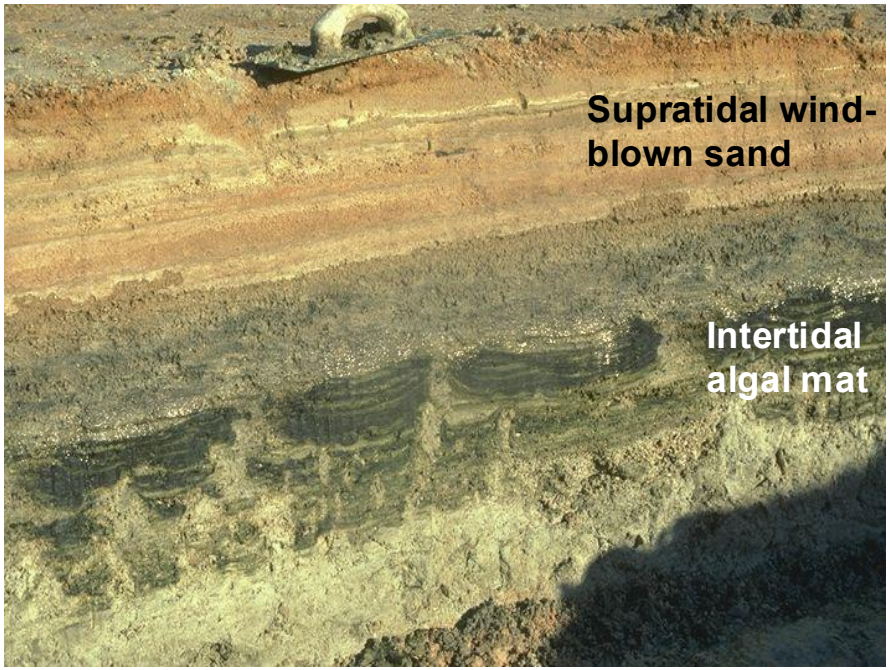


- Includes ponds and tidal channels
- High-diversity, high-abundance skeletal fauna
- Depositional processes and sedimentary structures:
  - Bioturbation
  - Sediment transport via tidal channels: fining-upwards successions, levee deposits



Peritidal facies – arid climate
Supra- and intertidal zones





- Depositional processes and sedimentary structures:
  - Halite crusts, gypsum, anhydrite
  - Dolomite
  - Algal mats
  - Desiccation cracks, teepee structures
  - Wind-blown/storm-deposited sand and mud
- Early diagenetic evaporite precipitation can disrupt/obscure depositional texture



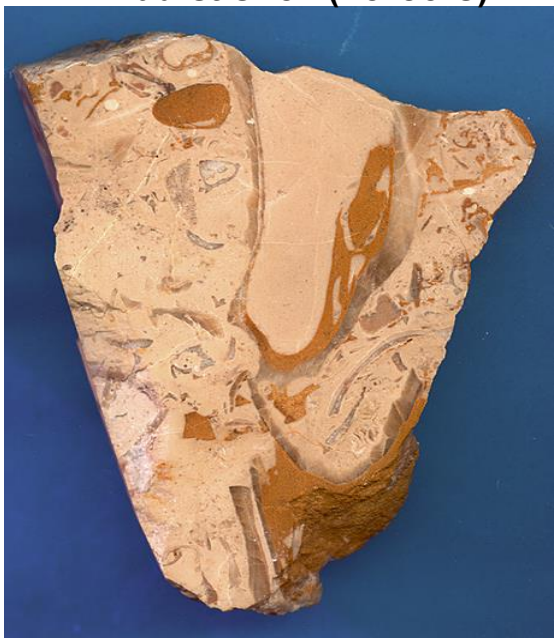
from Harris & Kowalik (1994)

Platform exposure

Karst

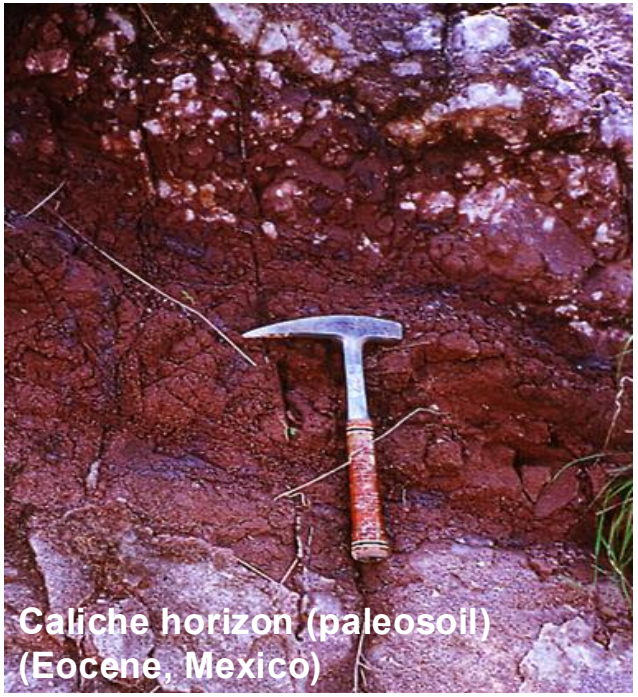


Aragonite dissolution in rudist shell (Vercors)





Root molds in Miocene limestone (Spain)



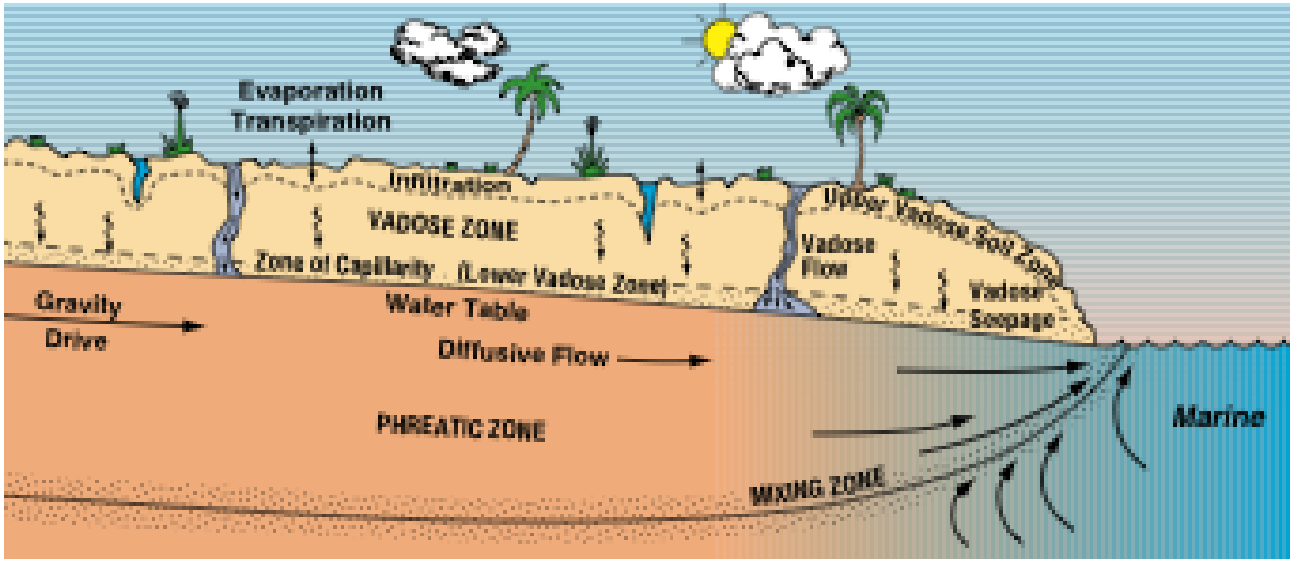
Caliche horizon (paleosoil) (Eocene, Mexico)

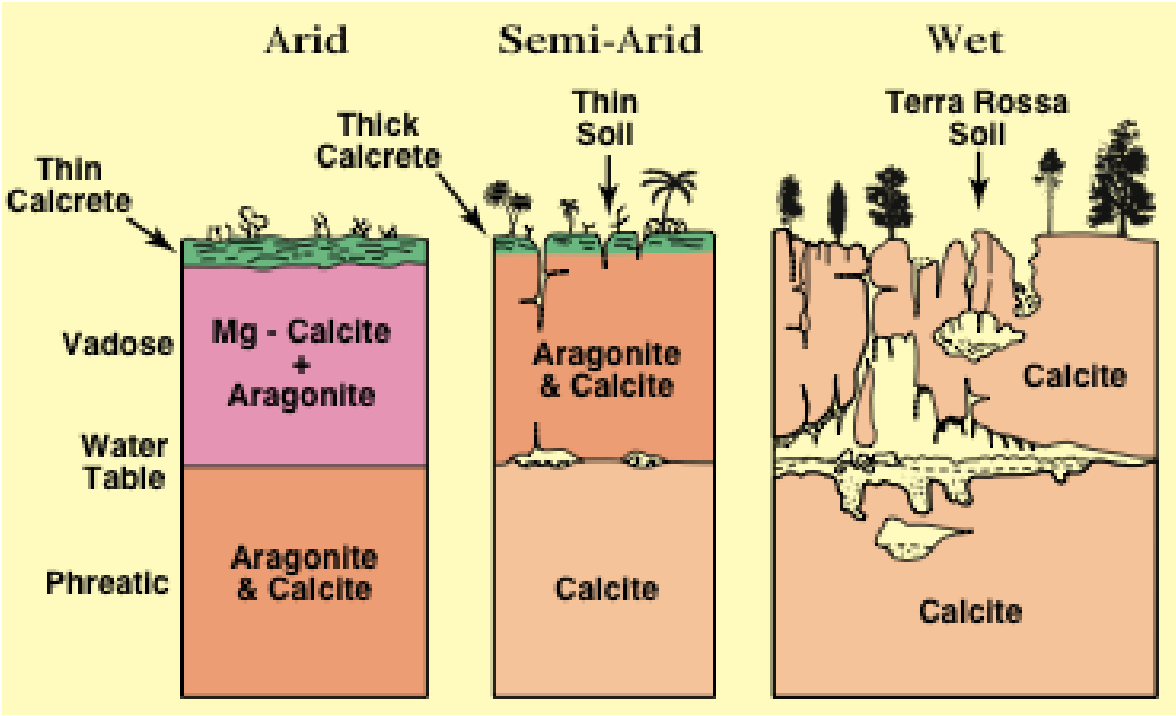
Photos: Arnaud

Platform exposure

- Rainwater reacts with soil to release CO_2 and form mild acid
- Downward percolation may follow pre-existing structural trends
- In humid setting create dense network of interconnected caves
- Subtle karst, calcretes in (semi) arid climate

Moore (1989)



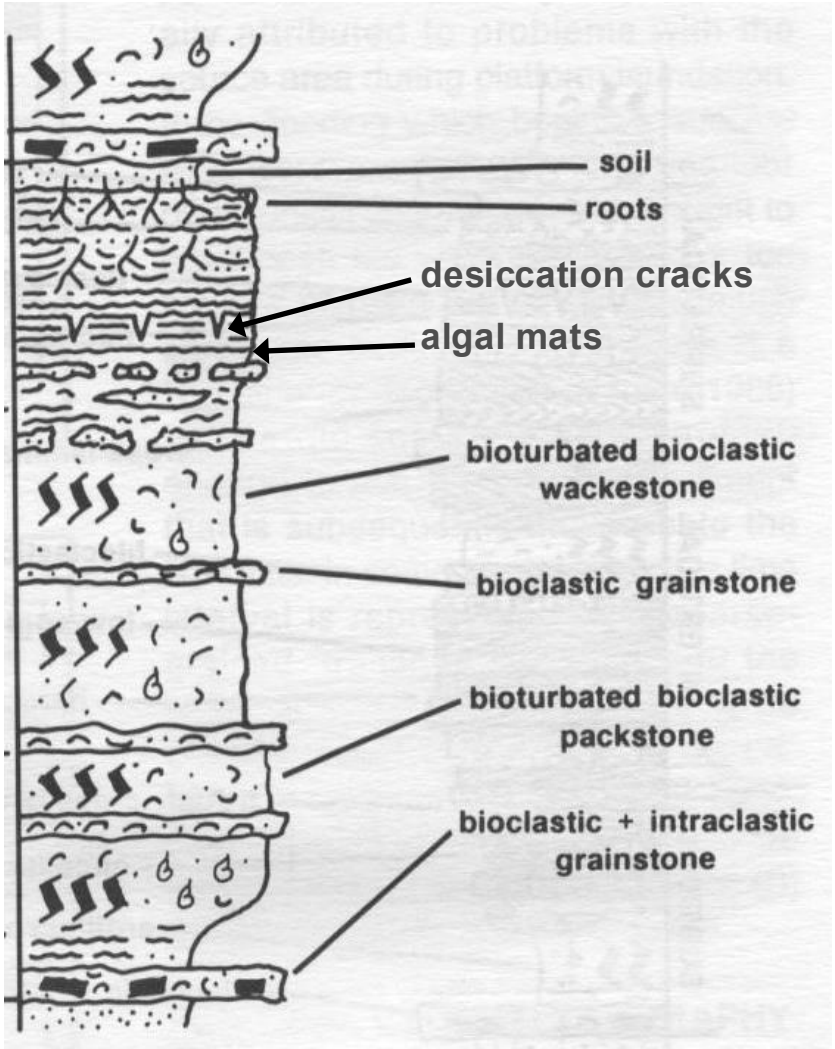


James & Choquette (1984)

Peritidal carbonates
Cyclicity

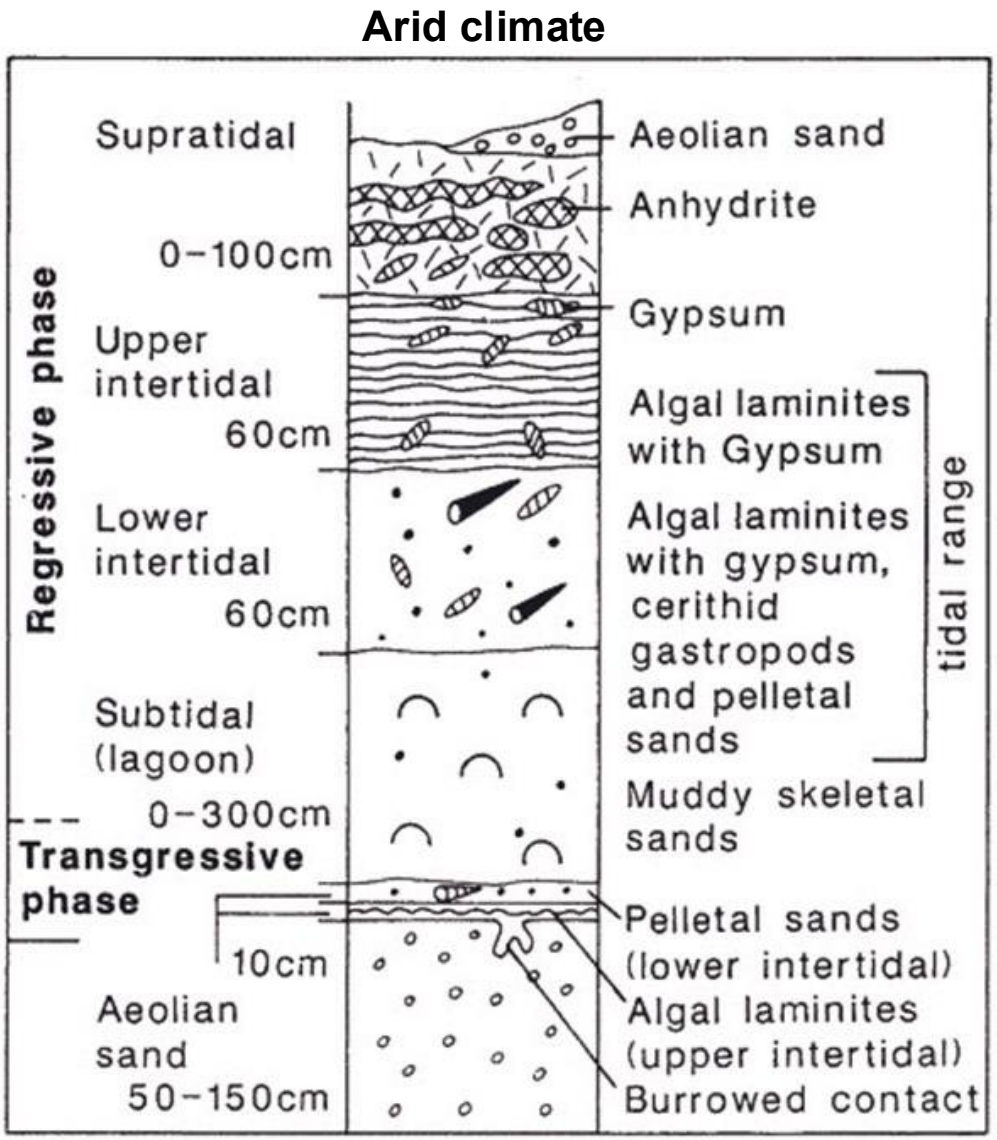


Humid climate

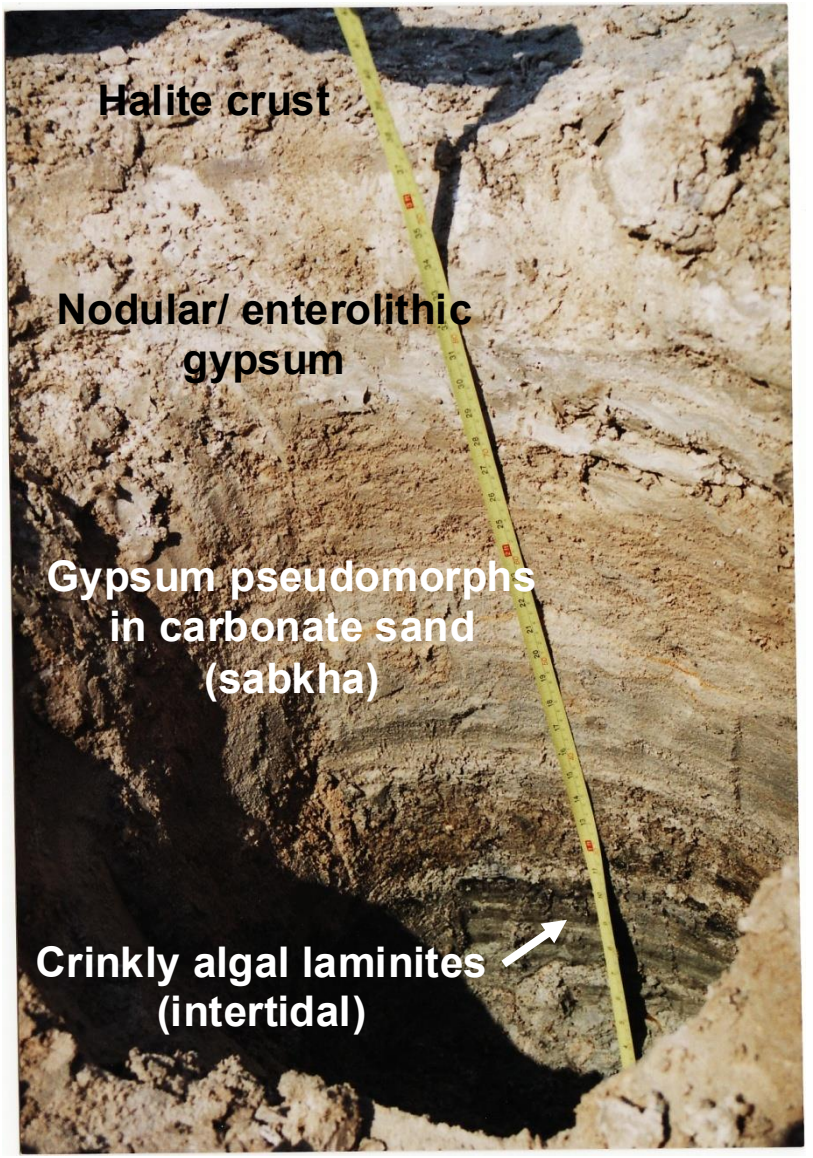


Walker (1984)

Peritidal carbonates
Cyclicity

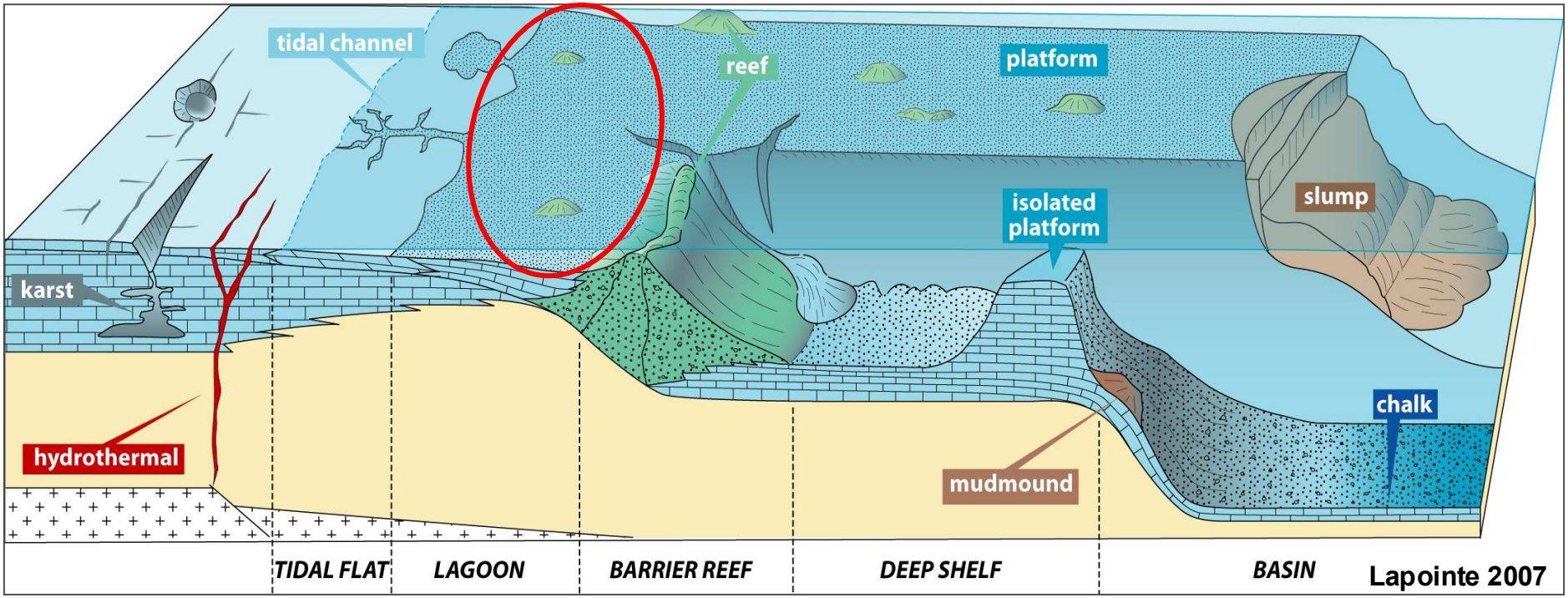


Tucker (1990)



Trench in sabkha, Recent, UAE

Carbonate depositional environments



Shallow subtidal environments
Lagoons and open shelf

- Moderate-/low-energy environments
- Muddy sediments, peloids, bioclasts (lime mud/wacke/packstones)
- Often pervasively burrowed
- Separation based on degree of connection to open ocean:

<u>Open shelf</u>

- No restriction, normal salinity
- Abundant & diverse fauna does not tolerate salinity variation (brachiopods, echinoderms, bryozoa…)

<u>Lagoon</u>

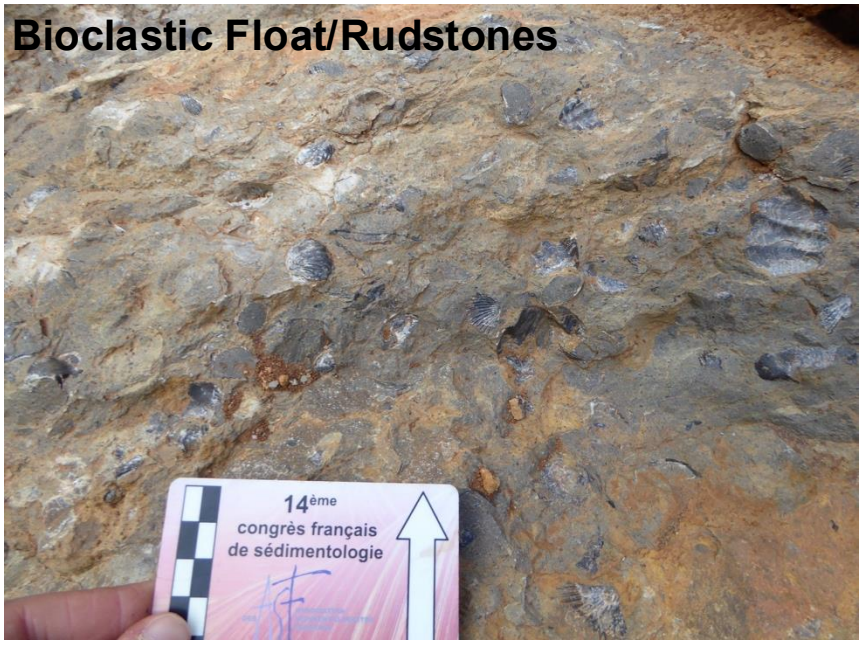
- Restriction, elevated salinity, storm recharge: salinity fluctuations
- Abundant but low-diversity fauna tolerates salinity variation (gastropods, algae, ostracods…)



Subtidal lagoon, Turks & Caicos



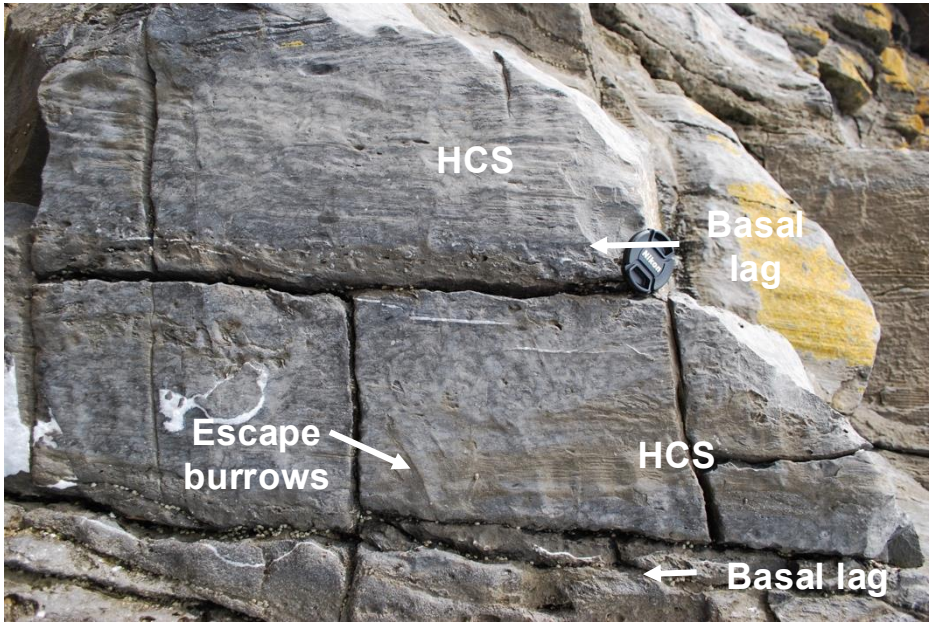
Shallow subtidal environments
Lagoons and open shelf





Bioturbation (*Thalassinoides*), Cretaceous, Oman

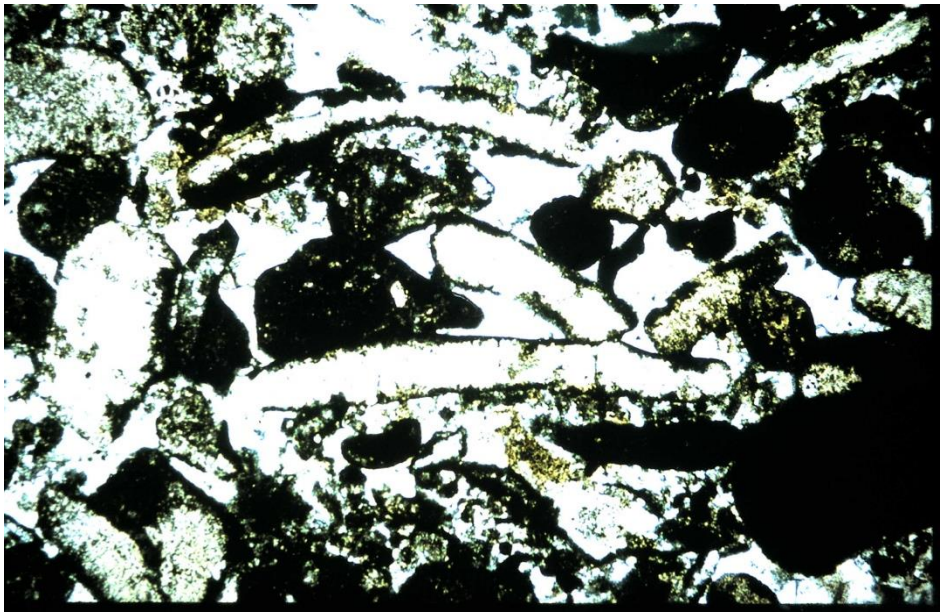




Three Cliffs Bay, Gower, South Wales

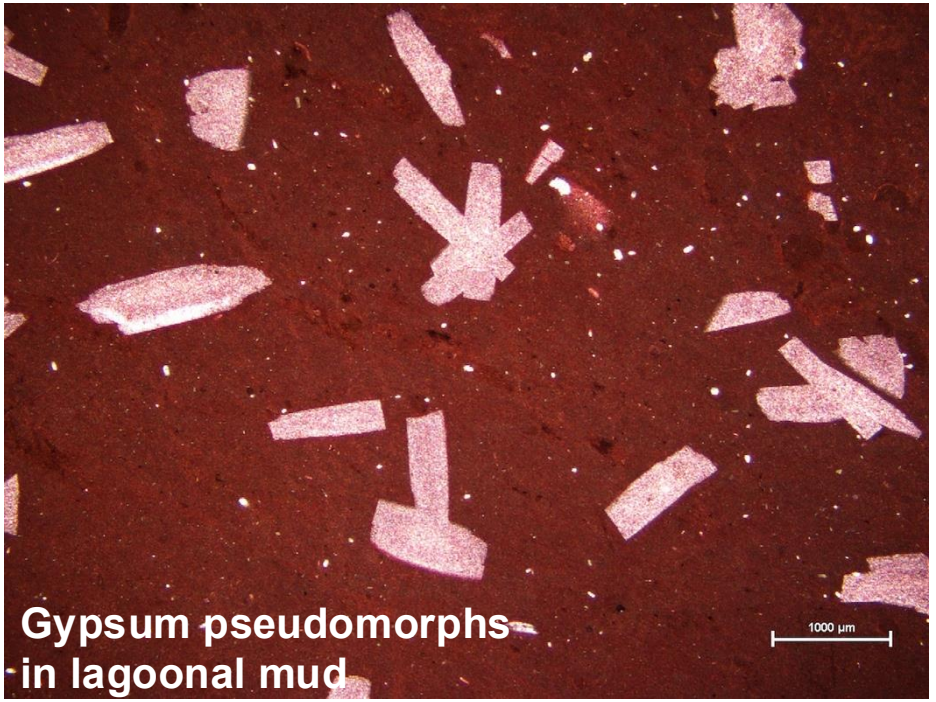
Shallow subtidal environments
Lagoons and open shelf

Micritization



1 mm

- Optimal conditions for fauna in shallow waters often lead to fossil-rich bioclastic limestones (float/rudstones)
- Fossil content will depend on salinity and substrate nature
- Micritization by cyanobacteria in illuminated shallow water



Gypsum pseudomorphs in lagoonal mud

Shallow subtidal environments
Carbonate sand bars/shoals



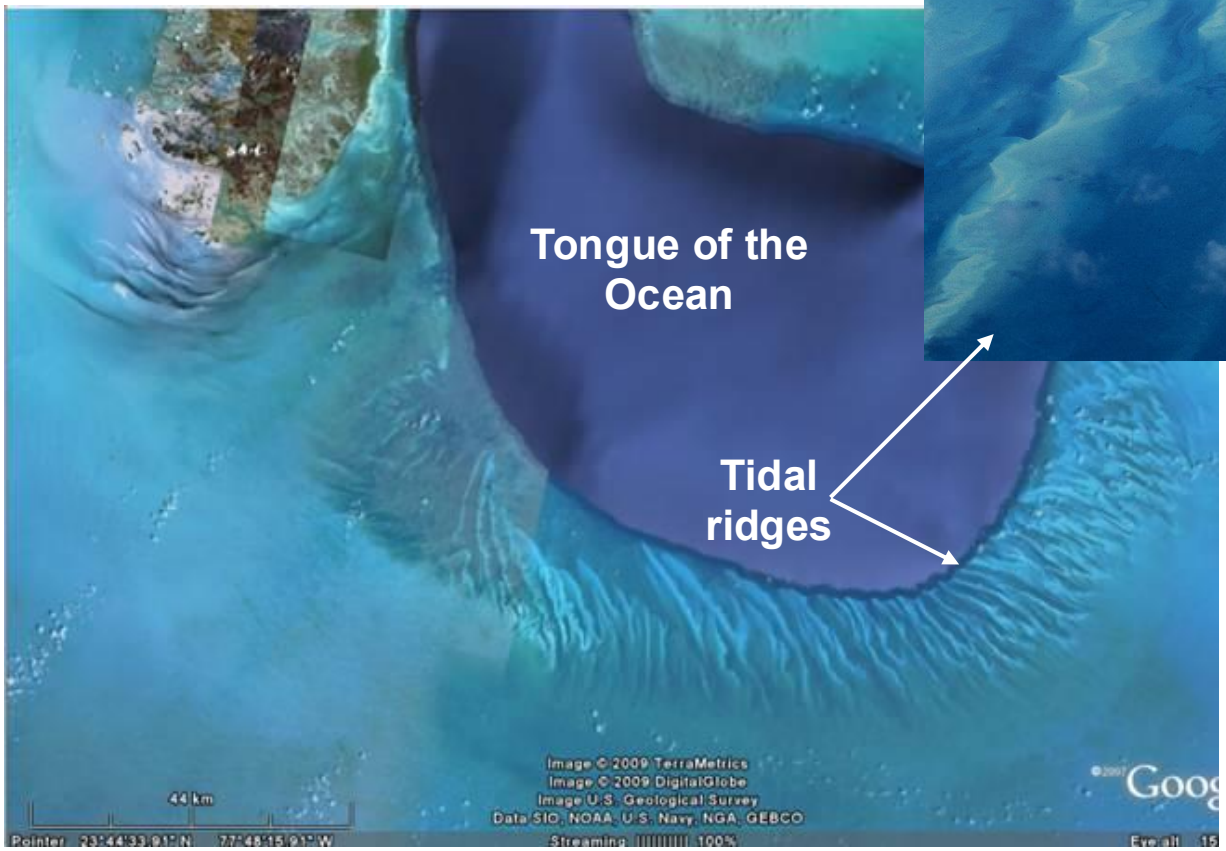


Harris & Kowalik (1994)

- High-energy & high carbonate production rates: shoreface, shelf edge, areas of strong currents (tidal, storms)
- Grainy sediments – important reservoirs
- Analogous to siliciclastic shorelines & sand waves

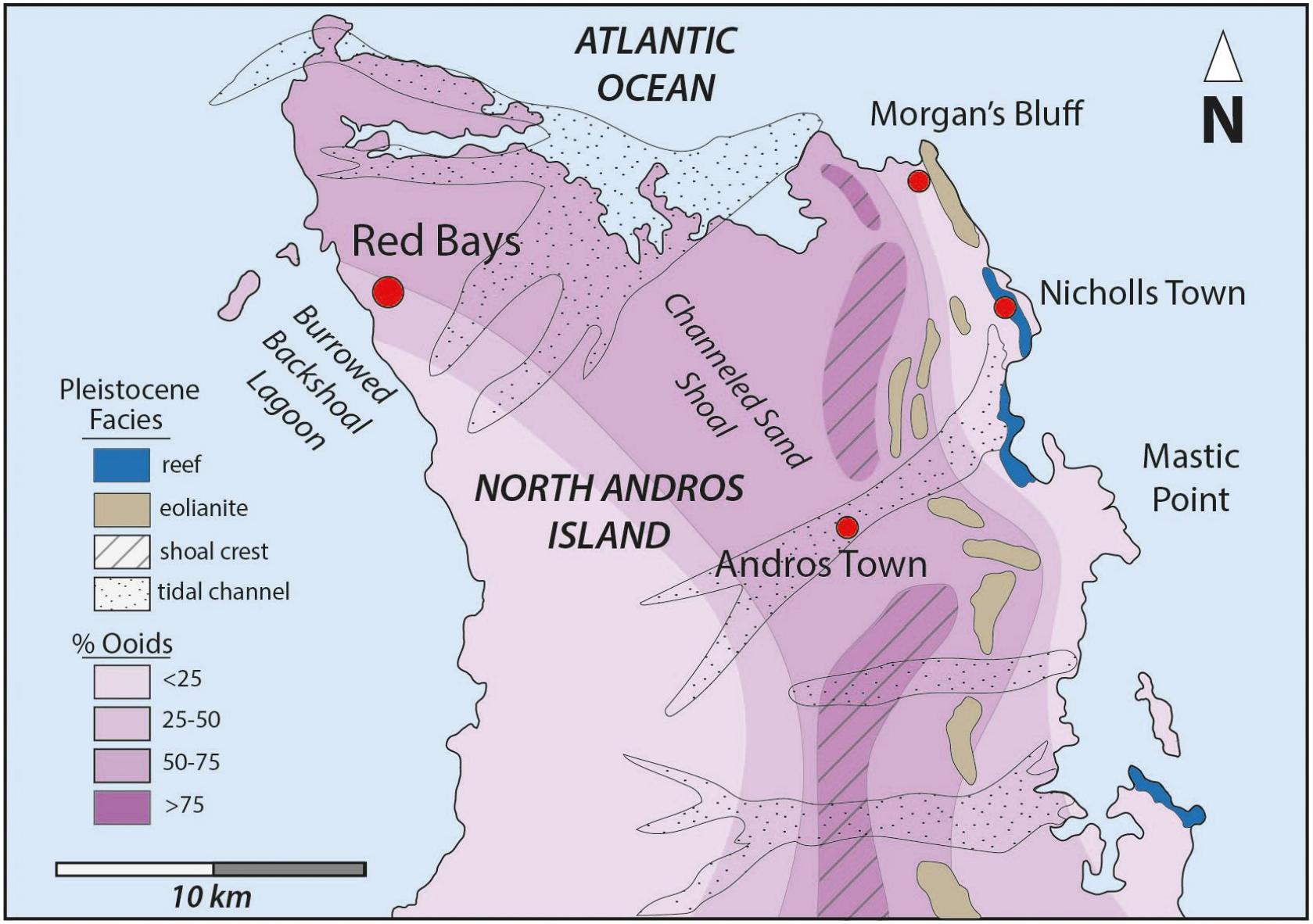
Shallow subtidal environments
Shelf-margin shoals





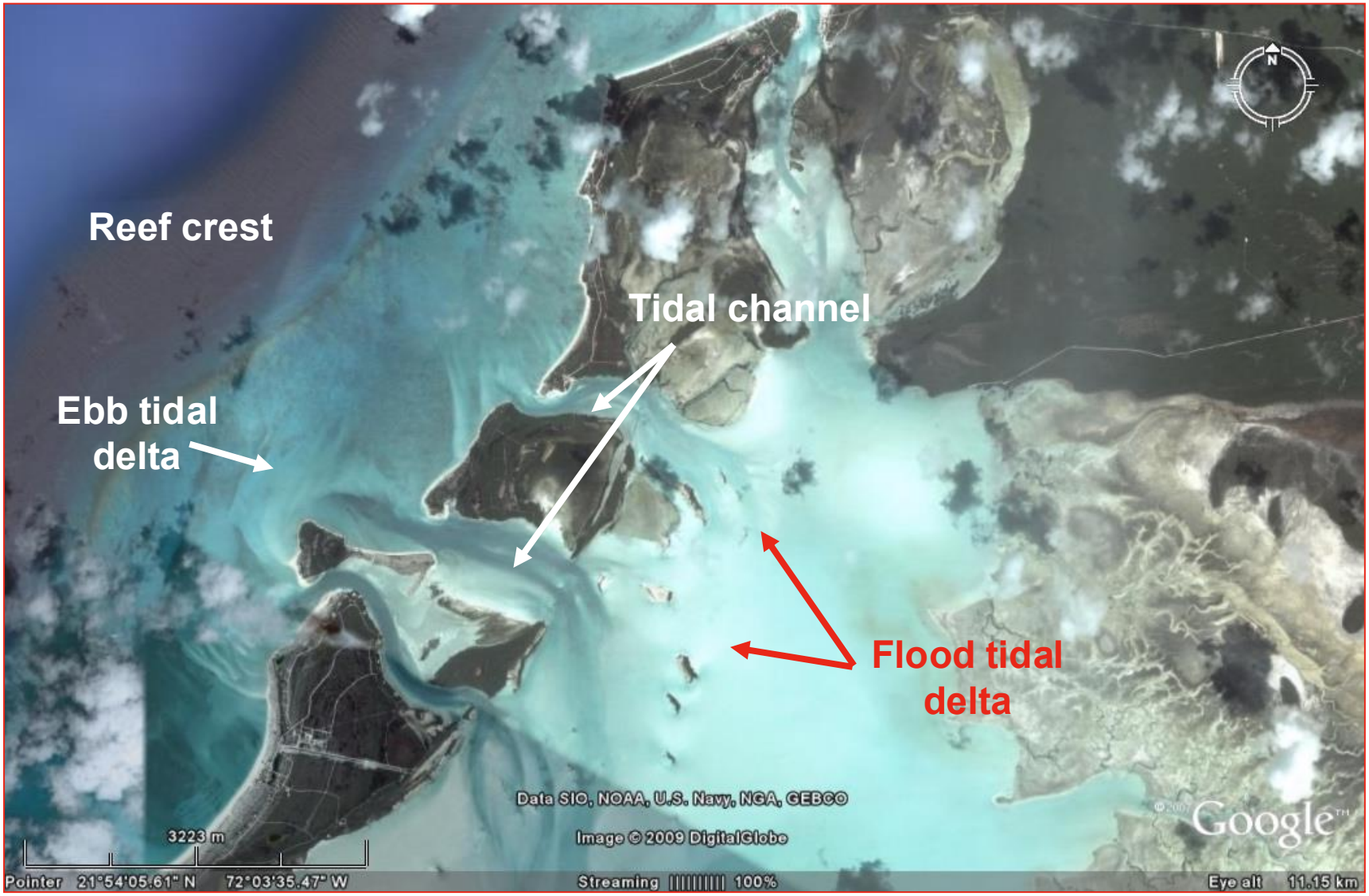
- Stable sand bodies, composed of oolitic and/or skeletal and peloidal carbonate sands
- Geometry and orientation strongly influenced by waves and tidal currents

Shallow subtidal environments
Shelf-margin shoals



Hazard (2017)

Shallow subtidal environments
Ebb/flood tidal deltas



Turks and Caicos Islands

Shallow subtidal environments
Shoreface

- Fairweather wave base to low tide
- Sand spits (longshore drift) and mobile sand shoals
- Beach-barrier island complex
  - Tidal channels
  - Stable, low-gradient coast
- Strandplain systems
  - Low/moderate topography, often rocky coastlines
- Depositional processes and sedimentary structures:
  - Wave-controlled sedimentation
  - High rates of carbonate sediment supply
  - Low-angle cross stratification in shoreface oolitic/skeletal grainstones
  - Early lithification: beachrock



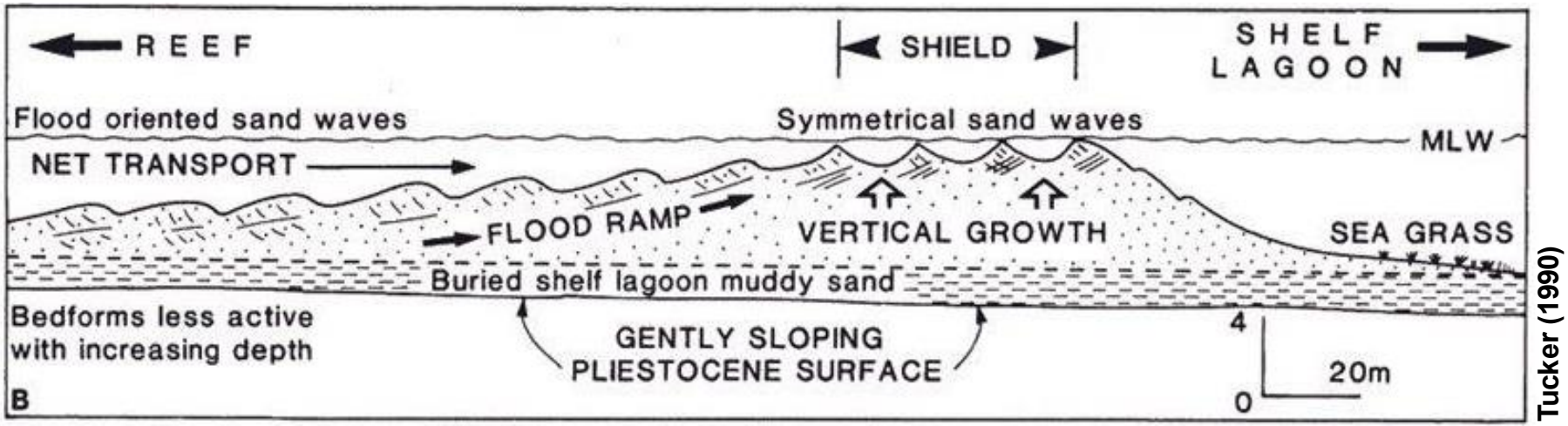
Quintana Roo, Mexico; William Ward/SEPM


Beachrock Cayman Islands

Shallow subtidal environments
Bedforms



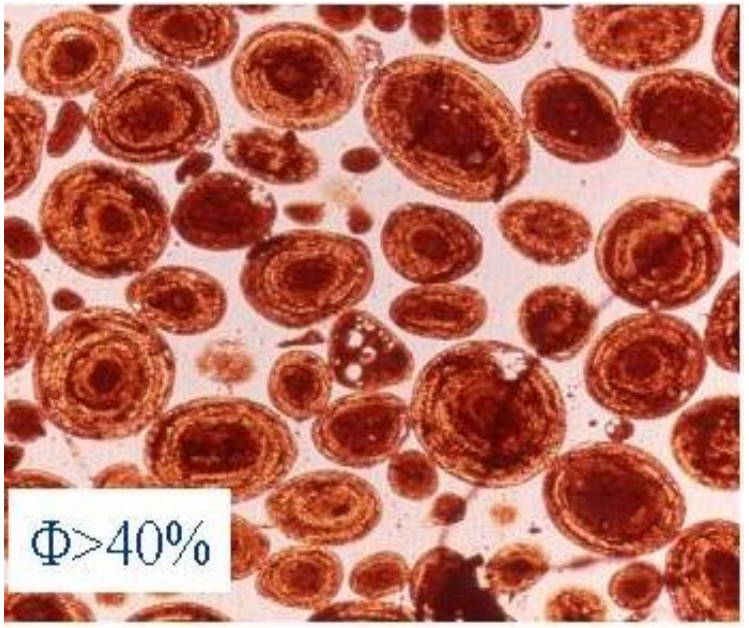




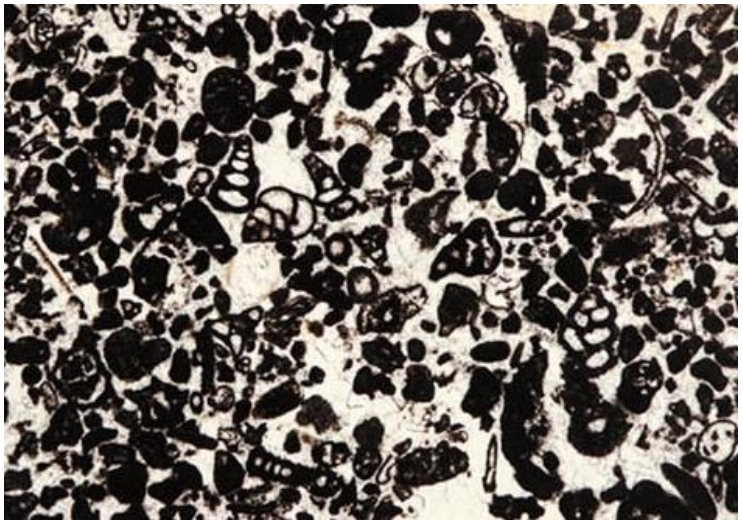
Tucker (1990)

Shallow subtidal environments
Carbonate sand bars/shoals

- Usually very clean, well-sorted carbonate sands
- Lateral and vertical heterogeneous: finer horizons, cemented beds (caliche, beachrock)







Summary

- **Peritidal carbonates form in depositional environments around the tidal range: from shallow subtidal to supratidal.**
- **Peritidal carbonates have characteristic sedimentary structures: algal mats, structures indicating partial exposure, and early diagenetic modification (roots, precipitation of dolomite and sulphate evaporites). Structures vary depending on climatic regime (humid vs. arid).**
- **Protected subtidal areas include lagoons and open shelves.**
- **Typified by low-energy deposition (but storm deposits possible) and mudstones to packstones, float/rudstones; high organic productivity, bioturbation; faunal content depends on connection to open ocean.**
- **Carbonate sand shoals occur in high-energy areas in the shallow subtidal: near the shelf edge, in the shoreface, and interior areas of strong currents.**
- **Shoals tend to be elongate structures analogous to clastic sand waves; include bedforms characteristic of strong currents.**
- **Reading: Tucker (1990), p. 81-85, 93-100, 101-137, 137-164**